

Ifty Mohammad Rezwan

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EDUCATION

- **North South University** Dhaka, Bangladesh
Bachelor of Computer Science & Engineering; GPA: 3.08
Specialization Courses: Computer Vision, Natural Language Processing, Machine Learning
September 2014 - March 2020

RESEARCH INTERESTS

- **Computer Vision:** Convolutional Capsule Networks, Object Tracking, Any Shot Object Detection, Uncertainty Based Deep Learning, Detection on the Edge, Medical Imaging
- **Natural Language Processing:** Resource Constrained Language Models, Word Embeddings
- **Multimodal Problems:** Optical Character Recognition, Complex Scene Text Recognition, Visual Question & Answering
- **Robotics:** GPS-Denied Navigational Path Planning, Robot Swarm

SKILLS SUMMARY

- **Languages:** Python, C, C++, JavaScript, Bash, Java, R
- **ML Frameworks:** Scikit-Learn, Pytorch, TensorFlow, Keras, MxNet, Darknet, Scipy
- **Data Science Libraries:** Pandas, Numpy, Caret, MLr, Dplyr
- **Visualization Libraries:** Matplotlib, Plotly, Seaborn, D3, Bokeh, Ggplot
- **Image Processing:** OpenCV, ImageIO, Pillow
- **Version Control:** GIT
- **Database Tools:** MySQL
- **Development Tools:** HTML, CSS, React, Flask
- **Robotic Libraries:** Robotic Operating System(ROS), Webots

EXPERIENCE

- **North South University** Under Dr. Shafin Rahman
Research Student
Feb 2021 - Present
 - **Object Detection:** Few, Zero and Any Shot Object Detection grounded in Deep Learning
- **Apurba-NSU Research Lab** Under Apurba Technologies & Dr. Nabeel
Research Engineer
July 2020 - December 2020
 - **Problem Description:** Develop a real time text recognition based solution for diverse sets of government documents under EBLICT without any predefined real world data.
 - **Synthetic Data Generator:** Developed an end to end synthetic data generating pipeline which takes in a list of words, fonts and size as prerequisite.
 - **Data Collection:** For the synthetic data generation pipeline, we collected a number of fonts which best represent our problem domain. To address the rich representation on the variation of words in the synthetic dataset we collected a few corpuses and articles from various subject domains, cleaned them and created a unique list of words.
 - **Noise Mapping Schemes:** As synthetic data is not enough to model real world noise, we explored numerous noise modelling schemes ranging from hand-crafted distribution based noise cocktails to generative adversarial network based noising solutions.
 - **Novel Bangla Character-Level Tokenizer & Model Fitting:** We introduced a new rule-based Bangla character-level tokenizer which outperforms all other previous Bangla character-level tokenizers. We explored various convolutional recurrent and transformer based models for the best fit of our data.
 - **Real World Evaluation:** To evaluate our model for real world deployment and performance we created four separate test sets and annotated them ourselves. These test sets had their own diverse set of noises ranging from old documents to recent documents.
 - **Model API & Flagger Models:** As our solution was a part of a bigger product which contained multiple models and APIs, we created a Rest Based API and deployed our model to production. Due to the necessity of sending various incoming images to various models after segmentation, multiple rapid flagger based classification models were required. We created a real time mobilenet based classifier pipeline to address this issue of multiple flaggers. We used our synthetic data generator pipeline to create datasets for these flaggers which were supplemented by our noise schemes to reflect real world images.
 - **Supervision:** Our work and methodology was verified by consultants from EBLICT, BUET and IIT,Dhaka University
- **North South University** Under Dr. Nabeel
Research Assistant
Jan 2019 - Jan 2020
 - **Computer Vision:** Developed a novel classification algorithm and optimized variants of others

- **Object Detection:** Developed an optimized edge based object detection algorithm and deployed it on a few IOT devices
- **Optical Character Recognition:** Explored OCR problems through the use of deep learning from a multi-modal perspective

• Self-Employed

Under Various Teams

March 2019 - Present

• *Software and Hardware based Solutions, Consultancy*

- **Product Deployment:** Created and successfully deployed a few software and hardware based solutions for a number of clients ranging from established companies to defense based institutions
- **Consultancy:** Worked extensively with various teams working on machine learning based solutions. These solutions range from data science based solutions to deep learning based products.

PUBLICATIONS

- **MixCaps: Capsules With Iteration Free Routing (Accepted, DICTA2020):** We proposed an iteration free routing algorithm with extendable mixed capsules. This brought about better hierarchical feature representations while being memory efficient. They performed better on complex datasets such as Cifar-10 and Cifar-100 than previous cnn and capsule based architectures. *Technology:* Python, Pytorch, Matplotlib, & OpenCV (September '20)

PROJECTS

- **SUCHONA: A Systematic Approach Towards Bangla Optical Character Recognition Using Synthetically Generated Data With Rigorous Evaluations(Computer Vision, Natural Language Processing, Optical Character Recognition):** (Work in progress) Created a big dataset of synthetic images from Bangla text which resembles real world images through custom and adversarial data augmentations. We then use this synthetic data to train different custom OCR models with various custom modules. We test these models on difficult real life datasets produced by us in various adversarial conditions such as different fonts, real world noise etc. We observe that through synthetic images alone, we can achieve around a minimum edit distance of 0.9477 with synthetic data only. *Technology:* Python, C++, Pytorch, OpenCV & (Jan '21)
- **IntelTrack: A real-time tracker for mouse based experiments(Computer Vision, Edge Computing, Visualization):** (Work in progress) A mouse tracker that works at real-time of around 20 frames per second on an average CPU that tracks the mouse and provides relevant graphical results for pharmaceutical experiments without any special library. *Technology:* Python, C++, Darknet, OpenCV, Matplotlib, Seaborn & Pandas (Jan '19)
- **Embedding Visualizers (Language Models, Word Embeddings, Visualization):** A Library that takes in tokens from Language models and visualizes the embedding vector of all words on a graph. Provides insights into the nature of knowledge from language models and word relations. *Technology:* Python, JavaScript, Bokeh, TensorFlow & PyTorch. (March '19)
- **Emotion Detection with various popular image classification architectures (Image Classification, Visualization):** Tested Imagenet winning architectures on the FER13 dataset. Provided insights into the confidence interval of various architectures on random images to see which architecture is better and why. *Technology:* Matplotlib, PyTorch, Python & OpenCV (March '19)
- **NimbleNav: An automated Global Mapping System for GPS-Denied Navigation Stacks(Rejected, SRSS2020):** An automated Global mapping system that provides a global map to robots from aerial images within a very short time. Normal approaches require the agents to traverse the unexplored terrain and provide a map which can culminate to loss of time and agents. We provide two mapping based solutions, one including basic threshold based image processing and a deep learning based segmentation approach for more complicated region. We test our hypothesis in a simulation based environment. *Technology:* Python, C++, ROS, Webots, Gazebo Matplotlib, OpenCV & Multi Robot Protocols (July '20)

EXTRA-CURRICULAR ACTIVITIES

• Competitive Machine Learning:

- TIC-HEAP Cirta Particle Classification Challenge: 11th out of 163 participants
- ICLR Workshop Challenge #1: CGIAR Computer Vision for Crop Disease: 33rd out of 822 participants
- SARAS endoscopic vision challenge for surgeon action detection: 66th out of 75 participants
- object-CXR: Automatic detection of foreign objects on chest X-rays: 33rd out of 57 participants

• Self-Teaching: At present I am trying to teach myself:

- Bayesian Statistics and Bayesian Machine Learning through several books. The uncertainty part of it is really interesting and can be exploited in deep learning.
- Causal Inference through Bray Neal's course and book.
- To build business intelligence and machine learning based solutions geared towards real life clients.

REFERENCES

- **Dr. Nabeel Mohammad** Associate Professor
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- **Dr. Shahnewaz Siddique** Assistant Professor
Email: shahnewaz.siddique@northsouth.edu North South University